

SynthMicro — Whole Blood Smear XAI Report

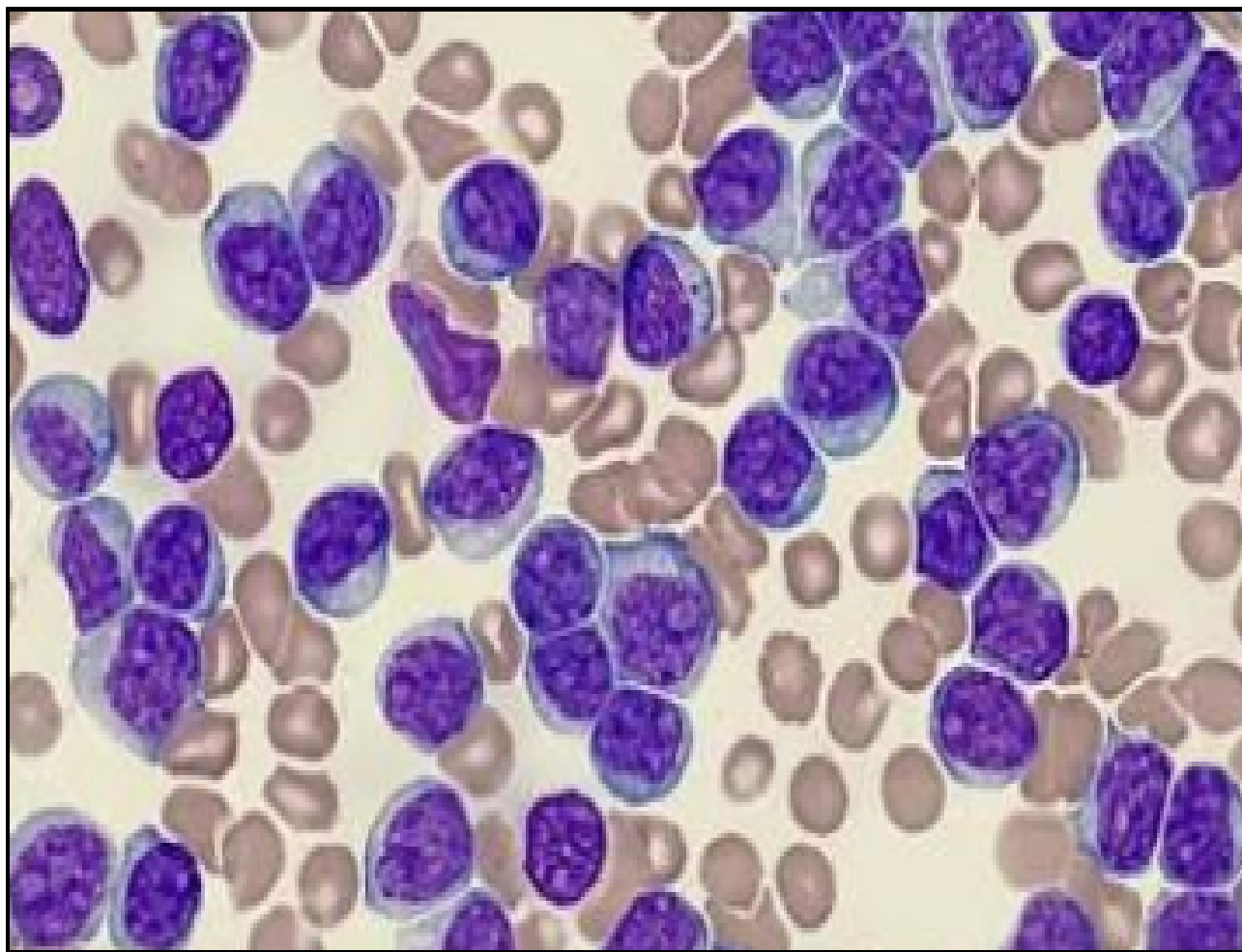
Analysis ID: 4c80692406df4781871ca1559a1f2156
Generated: 2026-09-07 21:11:35

Pipeline: Cellpose segmentation → 384×384 cell extraction → ResNet50 classification → Grad-CAM → SHAP → explanatory report.

Measurement	Result
Cellpose objects	49
Nucleated-cell candidates	48
Myeloblast-like candidates	17
Myeloblast-like proportion	35.42%

Screening interpretation: the proportion above is a research-oriented model indicator based on candidate cells selected by the notebook's purple-score rule. It is not a cancer probability and does not establish AML or another diagnosis.

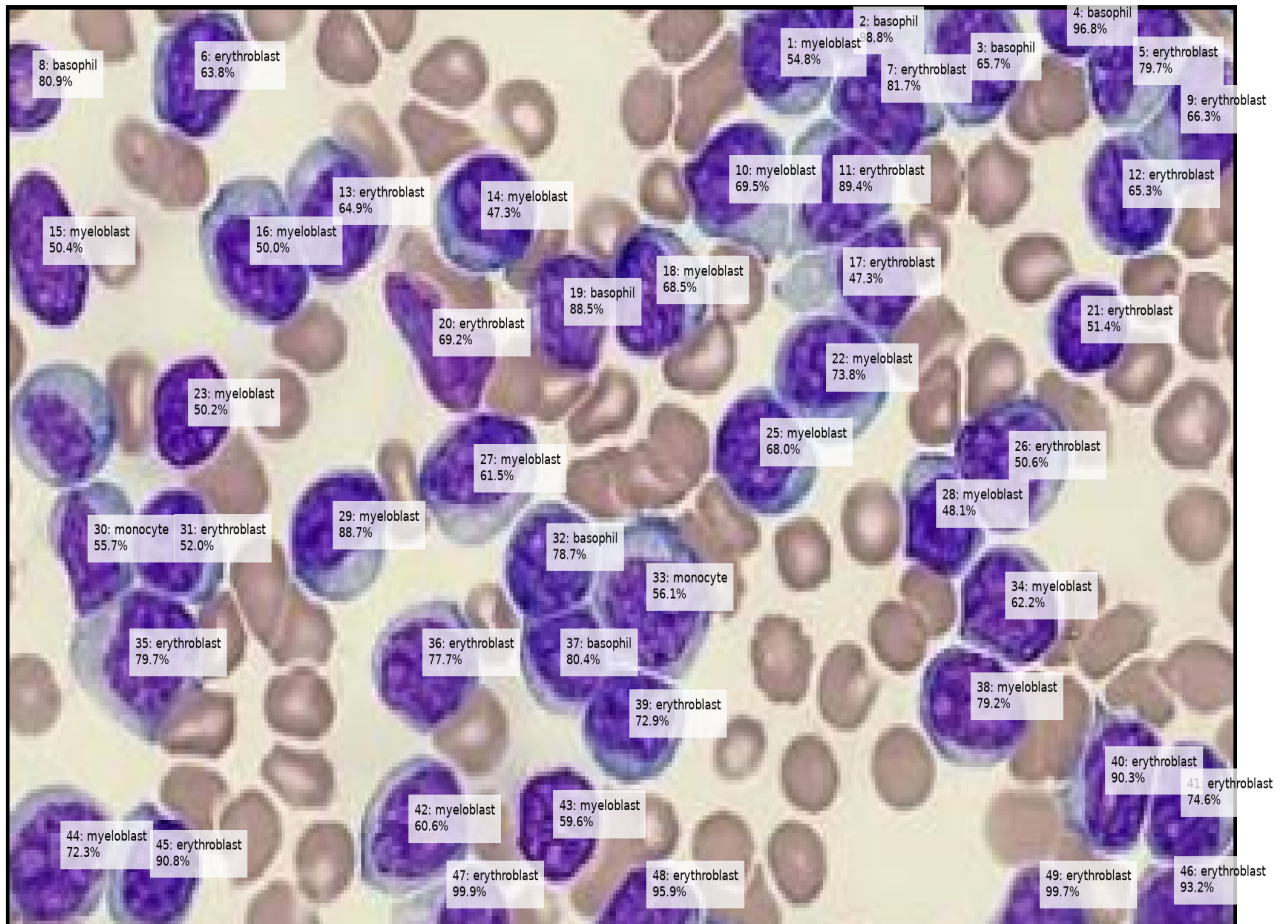
Whole Smear



Original uploaded blood-smear image analyzed by Cellpose.

Segmentation and Candidate Overlay

Whole Smear — Candidate Cell Classification



Cell-Level Results

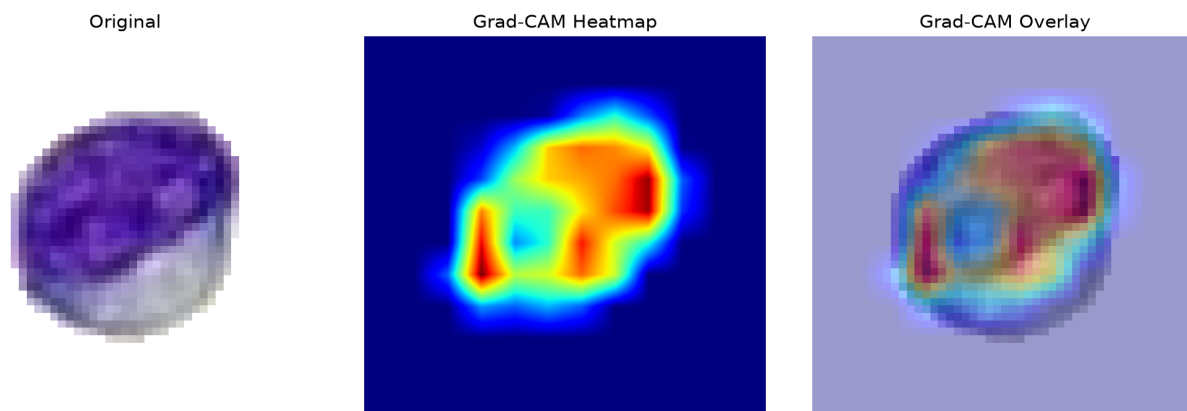
ID	Area	Purple	Class	Confidence
1	548	55.2	myeloblast	54.8%
2	218	66.8	basophil	98.8%
3	653	52.4	basophil	65.7%
4	171	68.9	basophil	96.8%
5	621	53.5	erythroblast	79.7%
6	659	56.8	erythroblast	63.8%
7	691	53.5	erythroblast	81.7%
8	274	49.8	basophil	80.9%
9	634	56.4	erythroblast	66.3%
10	749	48.0	myeloblast	69.5%
11	748	47.5	erythroblast	89.4%
12	570	55.7	erythroblast	65.3%
13	764	54.1	erythroblast	64.9%
14	678	56.2	myeloblast	47.3%
15	640	65.8	myeloblast	50.4%
16	837	52.3	myeloblast	50.0%
17	678	47.9	erythroblast	47.3%
18	693	54.2	myeloblast	68.5%
19	576	59.1	basophil	88.5%
20	571	62.6	erythroblast	69.2%
21	415	66.9	erythroblast	51.4%
22	799	48.9	myeloblast	73.8%
23	492	74.3	myeloblast	50.2%
25	664	56.9	myeloblast	68.0%
26	823	53.8	erythroblast	50.6%
27	821	47.0	myeloblast	61.5%
28	516	59.1	myeloblast	48.1%
29	736	51.6	myeloblast	88.7%
30	605	46.7	monocyte	55.7%
31	586	61.6	erythroblast	52.0%
32	593	53.9	basophil	78.7%
33	1029	46.6	monocyte	56.1%
34	663	59.6	myeloblast	62.2%
35	1001	46.0	erythroblast	79.7%
36	744	53.8	erythroblast	77.7%
37	511	57.8	basophil	80.4%
38	715	57.4	myeloblast	79.2%
39	649	52.4	erythroblast	72.9%
40	707	54.5	erythroblast	90.3%

ID	Area	Purple	Class	Confidence
41	612	61.7	erythroblast	74.6%
42	833	47.8	myeloblast	60.6%
43	537	68.6	myeloblast	59.6%
44	856	54.0	myeloblast	72.3%
45	607	55.6	erythroblast	90.8%
46	368	62.6	erythroblast	93.2%
47	325	43.4	erythroblast	99.9%
48	311	62.6	erythroblast	95.9%
49	259	57.3	erythroblast	99.7%

Grad-CAM Explanations

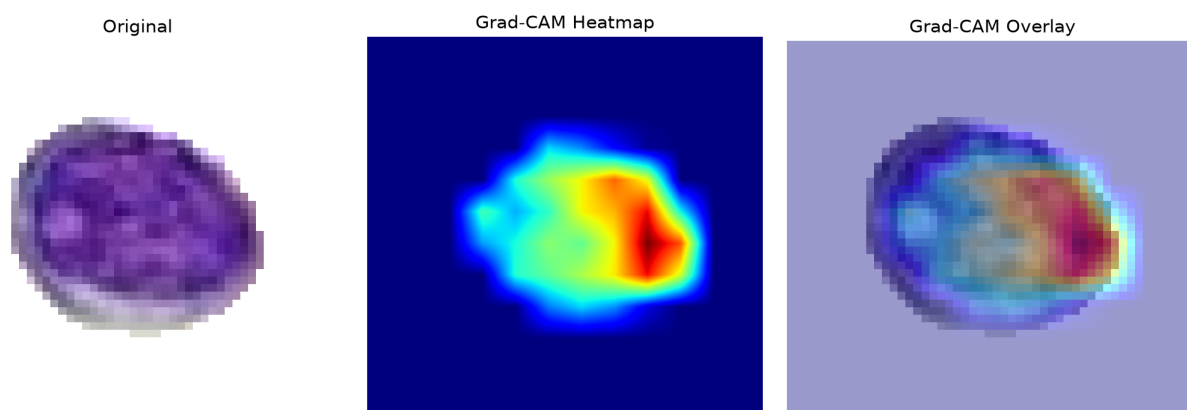
Grad-CAM shows regions contributing to the selected model class. It is an explanation of model behavior, not a direct measurement of a biological structure.

Cell 29 — myeloblast (88.7%)



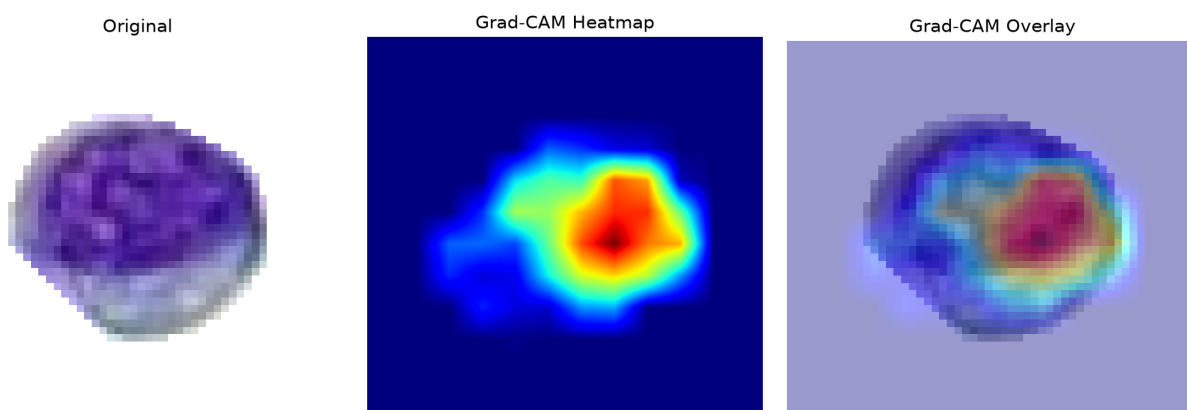
The ResNet50 model classified this segmented candidate as a myeloblast with 88.7% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 38 — myeloblast (79.2%)



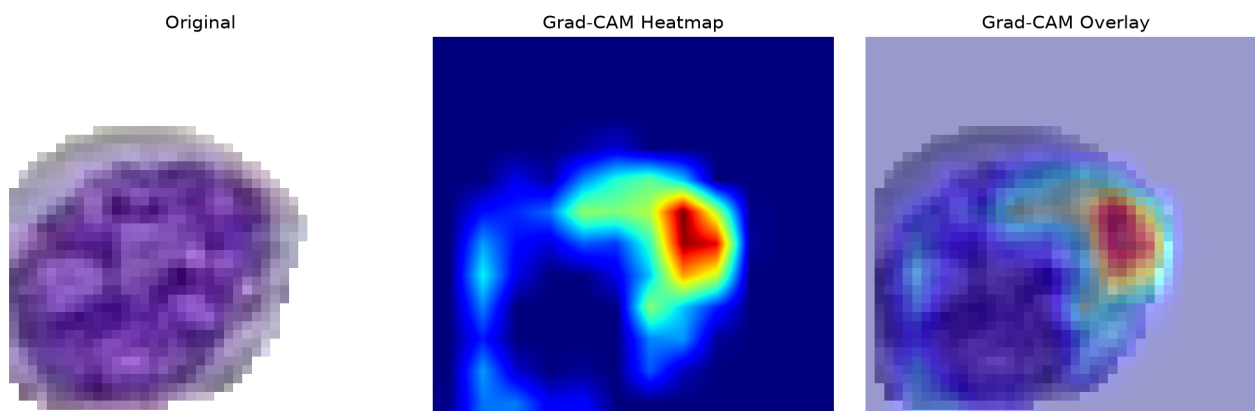
The ResNet50 model classified this segmented candidate as a myeloblast with 79.2% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 22 — myeloblast (73.8%)



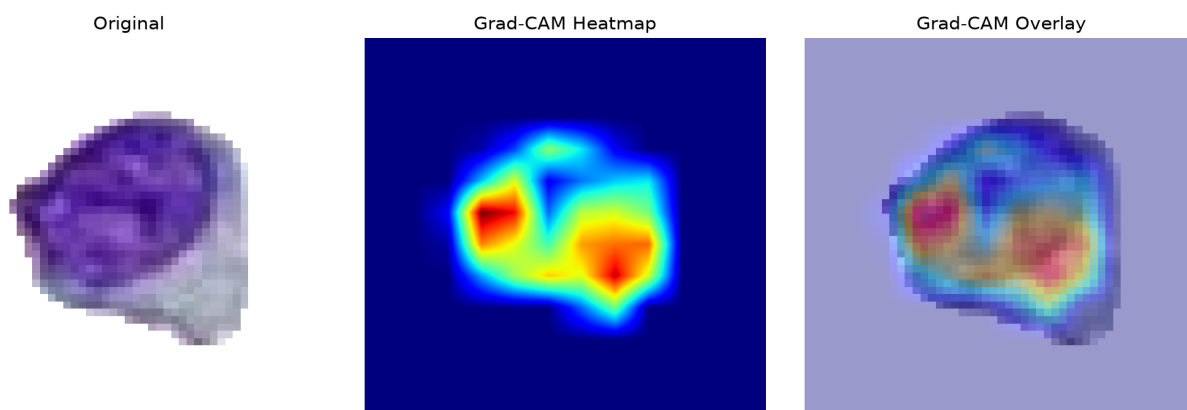
The ResNet50 model classified this segmented candidate as a myeloblast with 73.8% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 44 — myeloblast (72.3%)



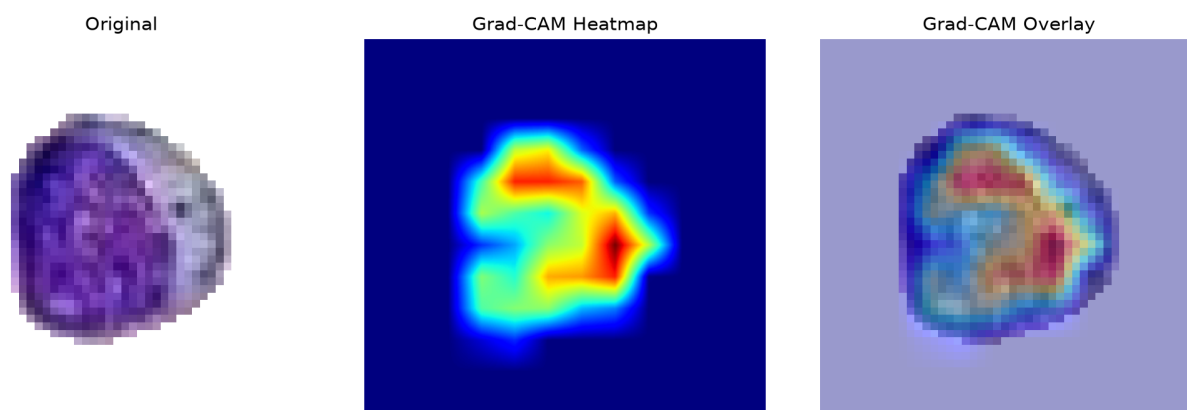
The ResNet50 model classified this segmented candidate as a myeloblast with 72.3% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 10 — myeloblast (69.5%)



The ResNet50 model classified this segmented candidate as a myeloblast with 69.5% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 18 — myeloblast (68.5%)



The ResNet50 model classified this segmented candidate as a myeloblast with 68.5% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

SHAP Explanations

SHAP attribution magnitude summarizes the magnitude of pixel-level contribution toward the selected class. It is a model explanation, not a clinical measurement.

SHAP output was unavailable for this analysis.

Interpretation and Limitations

- The classifier has five closed-set classes: basophil, erythroblast, monocyte, myeloblast and segmented neutrophil.
- RBCs and unknown cell types are not explicit classes and can therefore be forced into one of the five outputs.
- Cellpose segmentation quality affects every downstream result.
- Stain, microscope, illumination and acquisition differences can cause domain shift.
- Confidence is not a calibrated probability of disease.
- A myeloblast-like prediction is not equivalent to an AML diagnosis.
- Clinical assessment requires appropriate hematopathology and laboratory testing.